

OpenCloud Compute Console

Bangla Viva Answer & Personal Demo Guide

Sir-ke personally bujhanor jonno ready-made answer, demo flow, EC2 clone explanation, limitation handling, and likely follow-up questions.

1. Prothome Sir-ke Ki Bolben

Demo shurur age ৩০-৪৫ second-er moddhe project-er identity clear kore diben. Eta confidence build korbe.

Sir, amar project-er naam OpenCloud Compute Console. Amake EC2-er moto ekta cloud compute console banate bola hoyechilo. Ami eta AWS diye kori nai; eta Ubuntu Multipass-based EC2-inspired prototype. Ekhane EC2 console-er main workflow-gulo rakha hoyeche: dashboard, instances, launch instance, key pairs, images, security groups, volumes, network address, monitoring, billing, user role and settings. Current version-e UI, authentication, database, role permission, audit log and API structure ready ache. Instance workflow Multipass CLI try kore, CLI na thakle safe local fallback use kore. AWS-only kichu concept database/control-plane level-e represented.

Important line

Full AWS EC2 clone bolben na. Bolben: **EC2-inspired Ubuntu Multipass VM console prototype**. Eita honest and technically strong.

2. Sir Jodi Bole: “EC2 Clone Kothay?”

Short Answer

Sir, eta full production EC2 clone na, eta EC2 console-er core idea follow kore banano academic prototype. EC2-er moto compute resource management-er pages and workflows ekhane ache. Backend AWS-er bodole Multipass CLI target kore design kora.

Details-e Bujhaben

- EC2-te instance launch, key pair, security group, image, volume, IP, monitoring thake.
- Amar console-e oi same module-gular page, UI flow and API route structure ache.
- AWS API use korini, karon AWS proprietary platform. Instead Ubuntu Multipass CLI/daemon use korar plan.
- Multipass CLI thakle real VM action try kore; na thakle safe local fallback/unavailable state dekhay.

Sir, clone mane ami exact AWS copy kori nai. Ami EC2-er workflow clone korechi: instance manage, key pair, security group, image, volume, network address and monitoring. Backend architecture Multipass VM management-er jonno ready, tai eta local/private lab VM console hisebe kaj korte pare.

3. Main Objective-er Answer

Question: Ei project-er main objective ki?

Ei project-er main objective holo Ubuntu Multipass VM manage korar jonno ekta secure web dashboard banano. User login kore ek jayga theke instance, image, network, volume, security group, monitoring, billing estimate and activity logs dekhite parbe.

Kivabe bujhabe

Sir-ke dashboard open kore bolben: "Ekhane ami cloud console-er common modules group korechi. Left sidebar-e compute, network, storage, operations and administration section ache. Eta EC2-er moto resource management experience create kore."

Amar goal chilo ekta clean cloud management console banano, jekhane user technical API command chara web UI diye cloud resource manage korte pare.

4. AWS vs Ubuntu Multipass Answer

Question: AWS ar Ubuntu Multipass-er moddhe difference ki?

AWS holo Amazon-er public cloud provider. User Amazon-er infrastructure rent kore. Ubuntu Multipass holo lightweight local VM manager, ja nijer computer/server-e Ubuntu VM run korte help kore. Amar project AWS-er official API use kore na; eta Multipass CLI target kore.

AWS

- Managed by Amazon
- Public cloud
- EC2, S3, VPC etc.
- Proprietary service ecosystem

Multipass

- Lightweight local VM manager
- Lab/server environment-e useful
- Ubuntu VM, image aliases, networks and mounts
- University/lab/server environment-e useful

Sir, EC2 hocche AWS-er compute service. Ami oi compute console-er concept follow korechi, but backend-e Multipass VM command layer use korar architecture rekhechi.

5. Browser Theke Direct Multipass Call Na Korar Karon

Question: Browser theke sorasori Multipass call na korar karon ki?

Browser theke direct Multipass command call korle host access, command control, and secret expose hoye jete pare. Tai browser sudhu amar Next.js API route call kore. Server-side route authentication, permission, validation check kore tarpor Multipass layer call korbe.

Flow bolben

1. User browser theke button click kore.
2. Request jay local Next.js API route-e.
3. API route user session and role check kore.
4. Server-only Multipass module CLI/daemon call kore.
5. Normalized response UI-te return hoy.

Sir, security-r jonno ami cloud secret browser-e dei nai. Browser only amar backend-ke call kore. Backend gatekeeper hisebe kaj kore.

6. Auth.js, Prisma, PostgreSQL Keno

Question: Auth.js, Prisma, PostgreSQL keno use kora hoyeche?

Auth.js use korechi authentication and session management-er jonno. Prisma use korechi database schema, migration and query type-safe vabe manage korar jonno. PostgreSQL use korechi reliable relational database hisebe, jekhane user, role, settings, activity log and host connection data store kora jay.

Auth.js

Login, session, secure cookie, credentials provider.

Prisma

Schema model, migration, seed, safe database query.

PostgreSQL

Structured data, user-role relation, logs, settings.

Zod

Request data validation, invalid payload reject.

7. Role-Based Permission Kivabe Enforce Hoy

Question: Role-based permission kivabe enforce hoy?

Project-e three role ache: ADMIN, USER, VIEWER. Permission map define kora ache. API route-e requireApiPermission function current user-er session check kore, tarpor role onujayi permission ache kina verify kore. Permission na thakle 403 forbidden response dey.

Demo-te ki dekhate paren

- Admin login kore Users and Roles page dekhaben.
- User login kore dekhate paren admin-level action available na.
- Bolben: "Same app, but role onujayi feature access change hoy."

Sir, ADMIN user manage korte pare. USER resource action korte pare. VIEWER mainly read-only. Ei rule API level-e enforce hoy, sudhu UI hide kore security kora hoyini.

8. Password and Cloud Credential Security

Question: Password and host access secret kivabe secure?

Password database-e plain text hisebe rakha hoy na. bcrypt diye hash kore rakha hoy. Host access secret server-side thake, public environment variable-e rakha hoy na. Secret encrypt korar jonno AES-256-GCM helper ache, jeta IV, auth tag and encrypted payload use kore.

Extra security points

- Security headers middleware theke set hoy.
- Zod validation use hoy.
- Audit log safe message store kore, secret store kore na.
- Multipass modules server-only import diye protected.

Sir, amar project-e security layer multiple level-e: password hash, encrypted host access secret, role permission, API validation and audit log.

9. Current Limitation and Next Phase

Question: Current limitation ki?

Current version-e authentication, role management, UI shell, settings, activity log and API route ready. Instance workflow Multipass CLI call korar chesta kore, and CLI unavailable hole local fallback use kore. Limitation holo AWS-only feature-gulo, jemon true Elastic IP routing, EBS-style block volume, and firewall enforcement host automation chara fully real infrastructure-level na.

Next phase-e ki korben?

1. True historical metrics collector add kora.
2. Browser SSH terminal-er jonno safe websocket/session broker.
3. Host firewall automation diye security group enforce kora.
4. Multipass mount-based storage workflow aro production-ready kora.
5. Billing usage scheduler diye immutable usage record kora.

Sir, ami fake VM data dekhai nai. Multipass thakle real VM action try korbe, na thakle local fallback/unavailable state show korbe. Future-e host-level automation add korle aro production-like behavior pawa jabe.

10. Personal Demo Flow

Sir-ke personally bujhanor somoy ei order follow korle demo clear hobe.

1. **Login page:** bolben je project-e authentication mandatory. Demo admin: sakib09@gmail.com / 111111.
2. **Dashboard:** bolben eta cloud overview screen. Metrics cards and service status area ache.
3. **Sidebar:** Compute, Network, Storage, Operations, Administration groups dekhate hobe.
4. **Instances:** bolben EC2 instance list-er moto table/filter/action shell ready.
5. **Launch Instance:** bolben image, flavor, network, key pair, security group select korar flow.
6. **Security Groups/Key Pairs:** bolben EC2-er security model-er moto access control resource.
7. **Volumes/Network Address:** bolben storage and public IP management-er section.
8. **Users and Roles:** admin role diye user management dekhate hobe.
9. **User login:** rakib@gmail.com / 111111 diye normal user perspective dekhate paren.
10. **Limitations:** honestly bolben AWS-only infrastructure features host automation chara limited.

11. Sir-er Follow-Up Question Ready Answer

Q: Eta ki fully functional?

Full public-cloud production platform na. Eta academic prototype where UI, auth, database, permission, API architecture and core instance workflow ready. Host machine-e Multipass thakle VM action try kore, otherwise safe fallback use kore.

Q: Fake data keno dekhan nai?

Fake data dile cloud resource actually running mone hote pare. Tai ami honest placeholder rekhechi. Real API connect hole same UI live data show kore.

Q: Database keno dorkar, Multipass-e to data ache?

Multipass resource state Multipass-e thakbe. Local database user, session, role, settings, audit log, notification and encrypted host connection data store korar jonno.

Q: Project-er strongest part ki?

Strongest part holo secure architecture: browser direct cloud API call kore na, server-side API route, role permission, validation, password hashing, encrypted credential helper and audit log ache.

12. Final ৯-Minute Explanation

Sir, overall amar project holo OpenCloud Compute Console. Eta EC2-inspired cloud compute dashboard, but AWS-er bodole Multipass CLI target kore design kora. Current version-e ami UI shell, dashboard, instances, key pairs, images, network, volume, monitoring, billing, settings and user-role management-er structure ready korechi. Authentication Auth.js diye, database PostgreSQL and Prisma diye, permission role-based, password bcrypt diye secure, and future host access secret encryption-er helper ache. Browser direct Multipass call kore na; sob request protected Next.js API route diye server-side Multipass layer-e jay. Host machine-e Multipass CLI thakle real VM action try kore, na thakle safe fallback response dey. AWS-only kichu feature host automation chara limited, but architecture and demo workflow EC2 console-er moto ready.

Last tip

Sir-ke “complete AWS EC2 clone” na bole “EC2-inspired Multipass prototype” bolben. Ete apnar work realistic, honest and technically mature mone hobe.

Generated for OpenCloud Compute Console demo preparation.